Fernando Maceda Silva

Senior Software Engineer · Full Stack · Cloud & Distributed Architecture

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PROFESSIONAL SUMMARY

Senior Software Engineer with ~9 years delivering production systems across full-stack, cloud, and distributed architecture — spanning platforms used by 50+ enterprise customers (some exceeding 1,000 concurrent users) and a production SaaS platform serving live university deployments. Proven track record of technical decision-making: leading service extractions for independent scalability, driving database architecture choices, and cutting cloud costs 10–20% through infrastructure redesign. Combines deep hands-on engineering (JavaScript/TypeScript, C#/.NET, React, AWS/Azure) with a strong record of mentoring engineers and teaching software architecture, DDD, and full-stack development at both bootcamp and university level. Currently pursuing an MSc in Computer Science (AI specialization) and actively integrating AI-assisted development practices into team workflows, cutting rework by 50%.

TECHNICAL SKILLS

Languages & Frameworks: JavaScript, TypeScript, C#/.NET Core, React, Angular, AngularJS, Node.js, Python, Java, RxJS

Cloud & Infrastructure: AWS (Lambda, DynamoDB, SQS, CloudFormation), Azure (Functions, App Services, SignalR Service), GCP, Docker, Docker Swarm, Jenkins, Ansible

Databases: PostgreSQL, MySQL, SQL Server, DynamoDB, MongoDB, CosmosDB

Platforms & Tools: Rollbase / Infinite Blue (low-code PaaS), Workato, KendoUI, SignalR, Plop.js, Grafana

Practices & Architecture: Microservices, Event-Driven Architecture, Domain-Driven Design, REST APIs, CI/CD, SOLID Principles, Agile/Scrum, Technical Mentoring, AI-Assisted Development

PROFESSIONAL EXPERIENCE

Software Engineer

DMSi Software (via Jalasoft) · Omaha, Nebraska · Remote · Primary client assignment *Feb 2020 – Present*

Tech Stack: *JavaScript, TypeScript, AngularJS, KendoUI, Workato, Rollbase/Infinite Blue Platform, Grafana*

- Own and maintain two production systems on the Infinite Blue (Rollbase) low-code PaaS platform — a JavaScript customization layer and an embedded AngularJS/KendoUI widget — serving ~50 enterprise customer tenants, 10% of which exceed 1,000 users, within a 6-person cross-functional team.
- Designed and built a custom, platform-native release automation tool that replaced a manual XML-export, permissions, and post-upgrade process; cut release time from 30–120 minutes to under 5 minutes, now the team's standard release process.
- Diagnosed and resolved a critical performance issue for the platform's highest-volume customer, cutting page load times from 50–60 seconds to 1–2 seconds by eliminating redundant calculations.
- Led the architecture and integration of Workato as an event-driven replacement for in-house Outlook/Google integrations, improving scalability and offloading maintenance to vendor SDKs.
- Set up Claude Code (AI-assisted development) across the team, including custom platform-knowledge skills and a dynamic memory-generation pipeline from the app's XML schema; combined with new CI quality gates (linting, testing, TypeScript type checking) and shared coding standards, cut team rework by 50%.
- Fixed a critical widget outage affecting high-volume tenants by introducing role-scoped filtering and chunked pagination, eliminating a complete data-loading failure for large customer accounts.
- Independently researched and built a proof-of-concept for migrating the AngularJS widget to React, driven by AngularJS's deprecation and rising onboarding friction for developers with modern-stack backgrounds.
- Mentored 3 developers new to the team through onboarding, domain ramp-up, and code review; helped one engineer transition from an unfamiliar stack to full team productivity in under 2 weeks.

Software Engineer, Internal Product Team

Jalasoft · La Paz, Bolivia · Secondary engagement (parallel to DMSi) *Jan 2024 – Feb 2026*

Tech Stack: *C#/.NET Core, React, Azure (Functions, App Services, SignalR Service), MongoDB, SQL Server, SignalR*

- Served as technical lead and decision-maker for a virtual campus platform (video calls, chat, camera/mic/screen-share-based presence tracking) used by Jala University, evolving from an internal tool into a live production product, within a ~10-person engineering group.
- Partnered with UI/UX on feature scope and made data-architecture decisions across the stack — e.g., choosing MongoDB for high-write, flexible-schema features (polls, code snippets) and SQL Server for relationally-constrained features (tags tied to existing user/team relationships).
- Identified the chat module as a scalability bottleneck within a modular .NET Core monolith and led its extraction into an independent microservice, adopting Azure SignalR Service as a scalable event bus; now in production supporting ~1,000 concurrent users with reduced latency.
- Diagnosed and fixed a React rendering performance issue caused by redundant useEffect calls, applying code-splitting and memoization to eliminate visible UI lag.
- Drove infrastructure decisions during the team's Azure migration — replacing VM-based hosting with managed Azure services and shifting lightweight processes to consumption-based Azure Functions — cutting cloud costs 10–20% (tracked via billing dashboard) while maintaining performance.
- Transitioned from hands-on feature development to technical lead as the team scaled, guiding engineers in implementing designs and architecture decisions.

Software Engineer

Jalasoft · La Paz, Bolivia *Jul 2019 – Jan 2020*

Tech Stack: *.NET Core, Node.js, React, Docker/Docker Swarm, GCP, Grafana, Microservices, CI/CD*

- Contributed to a 3-microservice evaluations platform (templates and test instances with configurable question types/sections) alongside a React frontend, within a 10-developer team.
- Owned the Evaluations microservice, solving the core data-modeling challenge for organizing and moving questions between sections, and designing the associated API endpoints.
- Designed and built the team's deployment infrastructure from scratch: a dedicated infrastructure repository, a Docker Swarm-based multi-environment setup (dev/test/production) with proper inter-service communication, and a CI/CD pipeline enforcing a dev → test → production promotion flow.
- Contributed to the initial migration of infrastructure to GCP, setting up the deployment process on GCP virtual machines.
- Helped set up logging and alerting with Grafana, giving the team visibility into failed service deployments for faster manual remediation.

Software Engineer

Manheim (via Mojix) · Atlanta, Georgia · Remote *Apr 2017 – Mar 2019*

Tech Stack: *AngularJS, Angular, RxJS, Java, Oracle, AWS (Lambda, DynamoDB, SQS), Jenkins, Ansible, CloudFormation*

- Contributed to an automotive vehicle repair-and-auction platform (inventory, repair workflow, and auction sale) used by dealers, auction houses, and buyers, on a ~10-person team, working atop a Java/Oracle backend with business logic in stored procedures.
- Led the incremental migration of the frontend from AngularJS to Angular ahead of AngularJS's deprecation, introducing RxJS/Observables to modernize asynchronous data handling.
- Contributed to migrating business logic from an always-on Java process to on-demand AWS Lambda functions triggered via SQS (e.g., automatically resuming pending repair jobs when new inventory parts arrived), reducing compute costs by eliminating 24/7 execution for logic that only needed to run on events.
- Supported infrastructure operations during the AWS migration, learning to operate pre-built CloudFormation templates and Ansible playbooks to bring environments up/down for incident response.

PROJECTS

VS Code Extension — Infinite Blue Snippets

- Built a VS Code extension surfacing the Infinite Blue platform's server-side API directly in-editor, eliminating the need to cross-reference external documentation. Adopted by the developer's full team and by other teams working on the Infinite Blue platform.

Personal Portfolio & Technical Site

- Built a personal site (Hugo) showcasing skills, detailed experience write-ups, education, and personal projects.

TEACHING EXPERIENCE

Practitioner, Software Architecture 3

Jala University, Bolivia *Jul 2026 – Aug 2026*

- Designed practical use cases translating enterprise and microservices architecture concepts into applied exercises.
- Guided students through enterprise and microservices architecture design decisions; code-reviewed capstone projects.

Practitioner, Programming 6

Jala University, Bolivia *Jul 2025 – Sep 2025*

- Designed practical use cases for course technologies; guided students through implementation and code-reviewed capstone projects.

Instructor & Team Lead, Degree Project in Software Development 1 & 2

Jala University / Salesiana, Bolivia *Feb 2023 – Jul 2023*

- Designed course rubric per university standards; led teams of 8–10 students through a full product lifecycle (design through implementation) using Domain-Driven Design.
- Ran agile ceremonies as scrum lead; guided students through end-to-end problem-solving from design to implementation.

Frontend Instructor, Program DEV-33 Level 2 (Bootcamp)

Jala Foundation *Nov 2021 – Apr 2022*

- Designed a React/.NET Core curriculum from fundamentals through full-stack application development for a cohort of 10 students.
- Led students through UI design and implementation as part of agile cross-functional teams.

EDUCATION

Master of Science in Computer Science — *In Progress, Expected 2028*
University of York, United Kingdom · Specialization in Artificial Intelligence (independent research focus)

Diploma in Teaching for Higher Education — *2022*
Bolivian Catholic University “San Pablo”

Bachelor's Degree in Systems Engineering — *2017*
Bolivian Catholic University “San Pablo”

LANGUAGES

English: Professional Proficiency (B2–C1)
Spanish: Native
Italian: Basic (A1–A2)